

Self-Supervised Session Graph Recommendation with SR-GNN

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Abstract

We propose a lightweight self-supervised extension to session-based recommendation (SBR) with SR-GNN. Inspired by Self-supervised Graph Learning for Recommendation (SGL), a shared encoder aligns two perturbed session graphs alongside next-item prediction. We compare uniform node/edge perturbations with recency-aware perturbations that preserve the final observed item and target older context. Diginetica and RetailRocket experiments assess Recall@20, MRR@20, synthetic-noise robustness, training time, peak memory, and parameter count, testing recommendation and representation-robustness gains without substantially increasing model complexity.

1 Introduction

SBR predicts the next item from current-session interactions, especially for anonymous or first-time users without persistent profiles. Clicks on running shoes, socks, a water bottle, and a gym bag, for example, inform recommendations. Session-Based Recommendation with Graph Neural Networks (SR-GNN) models items as nodes and transitions as directed edges. Its gated graph encoder propagates item information into a session representation for prediction, capturing relationships difficult for purely sequential encoders (Wu et al., 2019). We retain this encoder and add SGL-inspired stability across perturbed graph views (Wu et al., 2021).

1.1 Usefulness

Short, incomplete sessions and noisy or incidental clicks can make representations overly dependent on individual items or edges. Consistency across perturbed views may encourage broader patterns rather than memorized transitions, improving prediction and reducing degradation under controlled noise.

1.2 Interest and scope

The project links graph neural networks and self-supervision without extra manual labels: does a collaborative-filtering objective also help SBR? We make one focused change to a well-understood

graph baseline; SASRec is only a non-graph evaluation baseline. No large language model, multimodal encoder, hypergraph, or extra prediction task is added. Self-supervised SBR already exists; our contribution is a controlled uniform-versus-recency-aware perturbation comparison for short, noisy sessions.

2 Potential Solution

2.1 Model and self-supervised formulation

We will reproduce or validate standard SR-GNN on a public SBR dataset. A session graph G yields representation z and next-item loss L_{rec} . Two augmented views G_1, G_2 share the encoder, producing a positive pair z_1, z_2 ; other sessions can provide negatives.

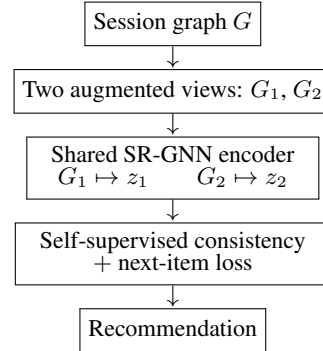


Figure 1: Shared-encoder pipeline: consistency between augmented views complements supervised prediction.

Following the SGL pattern, training minimizes

$$L = L_{rec} + \lambda L_{SSL},$$

Here L_{SSL} is contrastive consistency loss and λ is validation-tuned. Recommendation loss remains active throughout training; SSL supplements rather than replaces it.

2.2 Graph augmentations

Following SGL, uniform edge dropout removes a small fraction of transitions as the standard augmentation baseline; uniform node dropout is an ablation. Recency-aware perturbation always protects the final observed item, lowers removal probability for recent interactions, and perturbs older context more strongly. Views must retain at least

two observed items; augmentation is reduced or skipped for short sessions. The held-out target is never exposed or modified.

Semantic substitution using pre-computed item embeddings is a stretch goal only after core experiments, and only if both datasets provide reliable, comparable product metadata. This prevents a metadata encoder from confounding the main augmentation comparison.

2.3 Experimental design

We compare SASRec, standard SR-GNN, and SR-GNN with uniform or recency-aware SSL. Models share item vocabularies, chronological splits, evaluation prefixes, and comparable embedding dimensions. Ablations vary augmentation type/strength, SSL weight, temperature, and final-item protection. SASRec receives no SSL; it tests whether graph architecture and the auxiliary objective justify added complexity over attention-based sequence modeling (Kang and McAuley, 2018).

3 Datasets and Evaluation

We use two public e-commerce SBR benchmarks: Diginetika (approximately 204,000 sessions; 43,000 items) and RetailRocket (approximately 321,000 sessions; 51,000 items), under one documented preprocessing protocol. Their differing session lengths test dataset-specific effects (Choi et al., 2024). Development starts on Diginetika; the strongest configuration is then validated on RetailRocket, with identical preprocessing and evaluation across models.

We report Recall@20 and Mean Reciprocal Rank@20 (MRR@20), standard next-item metrics, plus training time per epoch, convergence time, peak GPU memory, parameter count, and optional inference throughput. We make no production real-time latency claim: two-view contrastive learning operates during training; inference uses one session and one encoder pass.

3.1 Robustness evaluation

Fixed noisy evaluation sets delete or perturb 5%, 10%, and 20% of observed interactions as runtime permits. The held-out target is never altered; every model receives identical corrupted sessions and random seeds. We compare absolute performance and relative degradation from clean-data baselines.

4 Expected Outcomes

Deliverables: reproducible SR-GNN/SASRec baselines; uniform and recency-aware view generators; auxiliary contrastive SSL; clean and controlled-noise experiments on both datasets; computational-cost measurements; and analysis of when SSL helps, fails, or does not justify training cost. Semantic substitution requires core completion and adequate metadata.

5 Timeline

Weeks 1–3: Review literature, audit data, finalize preprocessing, reproduce SR-GNN baseline metrics, add SASRec, and implement uniform and recency-aware view generation.

Weeks 4–5: Implement contrastive loss, validate the complete training pipeline, and tune SSL weight, temperature, augmentation strength, and final-item protection.

Weeks 6–7: Run controlled-noise robustness experiments, record computational costs, validate the strongest configuration on RetailRocket, and finish ablations.

Week 8: Finalize analysis, figures, report, presentation, and an optional demonstration.

References

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